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Dynamic Stochastic General Equilibrium and Business Cycles

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Research Track Master

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Chapter 4

New Keynesian Macroeconomics and Bayesian Econometrics

Content of the Handout

Handout Objectives

- Mastering the theoretical foundations of the New Keynesian framework;
- Understanding the relationship between VAR and DSGE models;
- Estimating a DSGE model using Bayesian techniques;
- Decomposing business cycles.

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1. The New Keynesian Model

The development of VARs models by Sims (1980) remains one of the major pathbreaking contributions for forecast and policy exercises as well as for business cycles analysis. The VAR methodology was able to overcome many identification problems faced by standard empirical methods and shed a light on interpreting macroeconomic time series and conducting monetary policy experiments. The VARs approach has gained broad acceptance and has allowed economists to establish how unexpected monetary policy changes lead to immediate effects on some macroeconomic variables but only slow, and hump-shaped, effects on others.

Following the empirical evidence of VARs regarding monetary policy, the standard RBC framework appeared to be irrelevant mainly because of the assumption that prices and wages were perfectly flexible. It became widely agreed that wages and prices did not adjust as quickly as needed to restore equilibrium, then monetary policy through, its refinancing rate, could be an effective tool to stabilize the economy and restore the best allocation.

In addition to this policy failure, the RBC model was also irrelevant to explain macroeconomic time series such as real GDP and hours. For instance, Gali (1999) pointed out an important anomaly in the RBC theory regarding the high positive correlation between hours and labor productivity which was generated by the model but not observed in the data. Departing from the flexible price hypothesis and introducing monopolistic competition into the RBC framework was able to fix the anomaly by obtaining an artificial correlation consistent with the data.

Sticky prices are the key feature of New Keynesian models. Combined with a monetary policy rule, this framework was successful in providing macroeconomic stability until the financial crisis episode in 2007. The New Keysenian framework is the current workhouse for the conduct of monetary policy. Policymakers can successfully estimate the New Keynesian model, and run policy exercises to find the best way to implement monetary policy, given the social preferences of households and the shape of the supply curve.

In this lecture, we concentrate on the compact version of the New Keynesian model, which can be summarized by 3 structural equations. These equations are micro-founded, *i.e.* derived from microeconomic behavioural equations, and remain fairly consistent with most of the Old-Keynesian theory. In particular, the framework includes an AS (*Aggregate Supply*) curve which is a combinason of the Phillips curve and Okun's law in a rational expectations equilibrium. The lecture also introduces to Bayesian econometrics applied to the 3-equation model for the US economy.

1.1 Empirical evidence

Before starting with the theoretical foundations of new Keynesian models, it is worth focusing first on the empirical evidence regarding the macroeconomic interactions between real output variations, inflation rate and nominal interest rate through a VAR model. In this section, we use a BVAR model (Bayesian Vector Autoregressive): the difference with standard VAR models lies in the fact that the model parameters are treated as random variables, and prior probabilities are assigned to them. Bayesian methods are also employed for the estimation of DSGE models. Given the good empirical performances of BVAR models, they usually constitute a benchmark that DSGE models aims at replicating in terms of forecasting performances and dynamic properties.

The New Keynesian framework aims at giving theoretical foundations for the macroeconomic links between real output, the inflation rate and the nominal interest rate. It is thus possible to estimate a BVAR(4) model for these 3 times series. To remove trends, real output is *per capita* and in logs, then we apply a first difference filtering to get the growth rate. Dynare incorporates routines for BVAR models estimation, that can be used alone or in parallel with a DSGE estimation. Our set of observable variables is composed of the growth of real output Δy_t^{obs}, the inflation rate π_t^{obs} and the FF rate r_t^{obs} for the US

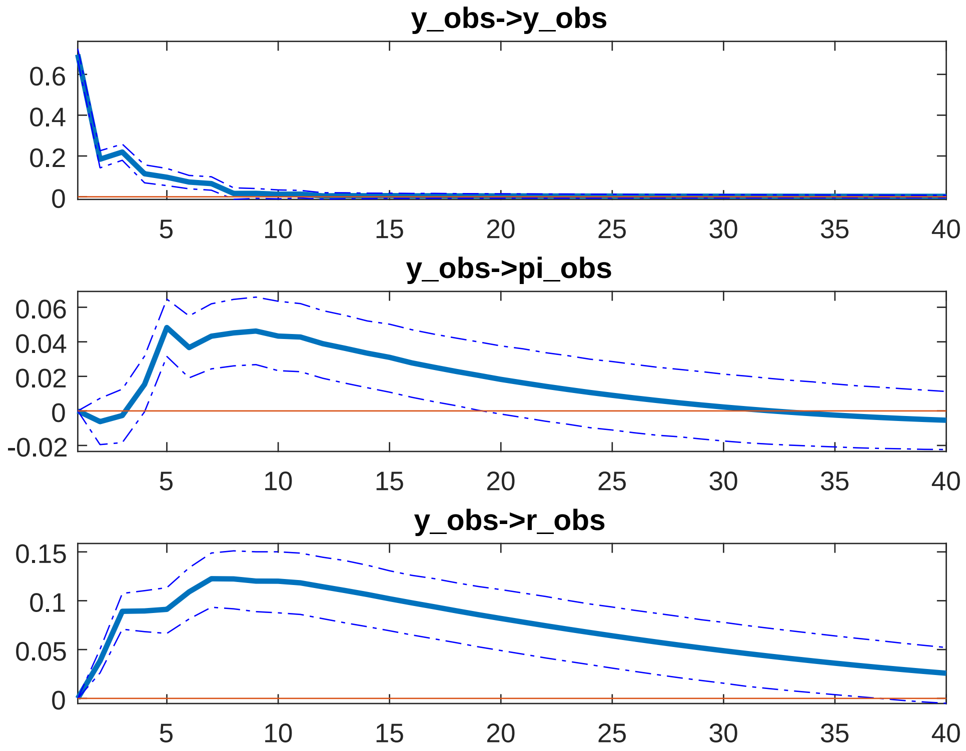


Figure 1: BVAR IRF after a shock on the real GDP growth rate equation (first panel is Δy_t, second is π_t and third is r_t).

economy between 1970 and 2015 on a quarterly basis.

Figure 1, Figure 2 and Figure 3 deliver the response of the model. Figure 1 can be fairly associated to a demand shock characterized by an exogenous increase of output, followed by a rise of both the inflation and the interest rate. Figure 2 reports the response of the system after a shock on the inflation rate equation. The growth of output tends to decrease while the interest rate increases. The opposite sign of response of output and inflation characterises a supply shock. Finally, Figure 3 reports a shock to the interest rate equation. This shock strongly depresses output while it increases the inflation rate. The second result is rather surprising as a federal fed rate hike is expected to lower the rate of inflation through a standard IS-type demand effect. Rabanal (2007) answers this question through a DSGE model while Christiano et al. (2005) incorporate the interest rate in the marginal cost of goods so that a monetary policy tightening increases inflation through the marginal cost of goods.

The Dynare code for a BVAR reads as follows:

code

```
var y_obs pi_obs r_obs;
varobs y_obs pi_obs r_obs;
bvar_density(datafile=data_us_nk,xls_sheet=obs,
             presample=15, prefilter=1, first_obs=1,nobs=182,
             xls_range=B1:D183, noconstant) 4;
bvar_irf(4,'SquareRoot');
```

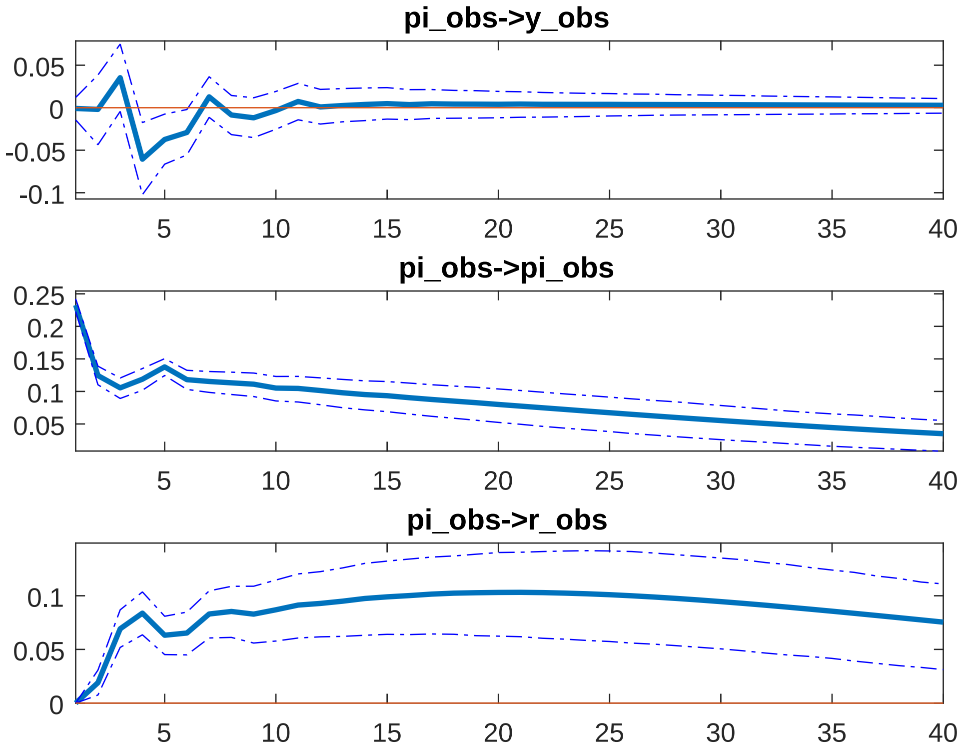


Figure 2: BVAR IRF after a shock on the inflation rate equation (first panel is Δy_t, second is π_t and third is r_t).

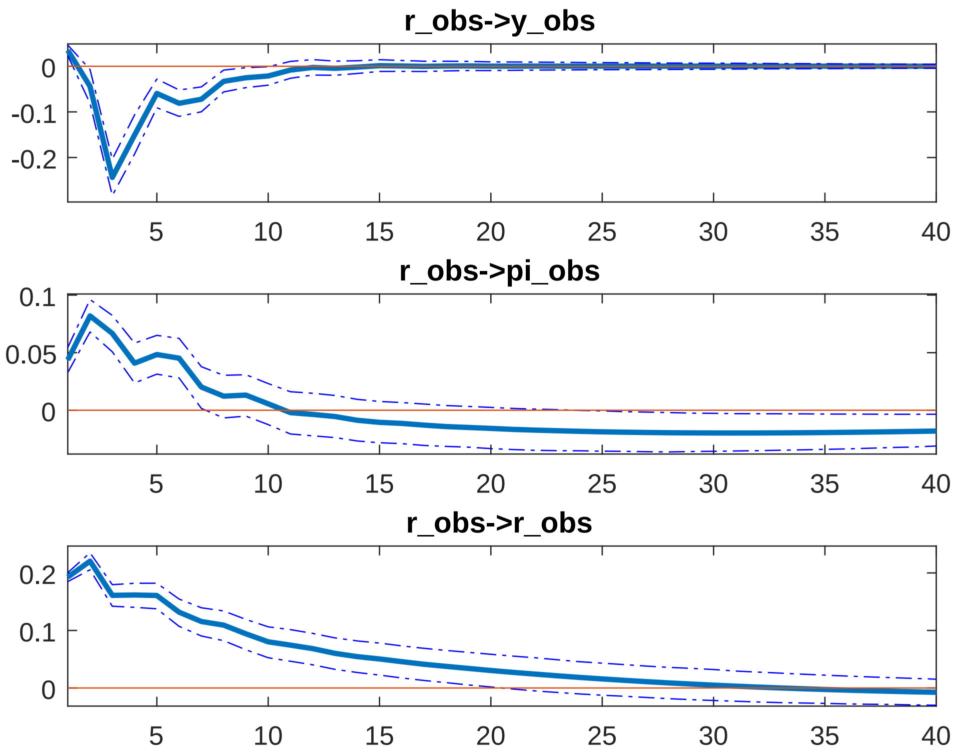


Figure 3: BVAR IRF after a shock on the nominal rate equation (first panel is Δy_t, second is π_t and third is r_t).

1.2 Theoretical foundations

In a Nutshell, the conventional New Keynesian model can be summarized with a system of three log linear relations that determine three main variables of interest:

- the output gap ($\hat{y}_t$),
- the inflation rate ($\widehat{\pi}_t$),
- the nominal interest rate ($\hat{r}_t$).

Although this model is now standard, we present it in some details, to fix the notation and the logic underlying the structure of each relation. In the New Keynesian model, each period represents a quarter.

1.2.1 The New Keynesian Phillips Curve

The first relation is the New Keynesian Phillips Curve (NKPC) that links current inflation ($\widehat{\pi}_t$) to the expected inflation ($\mathbb{E}_t\widehat{\pi}_{t+1}$), the output gap ($\hat{y}_t$) and to an exogenous supply shock (ε_t^S) that takes the form of a cost-push shock:

$$\widehat{\pi}_t = \beta\mathbb{E}_t\widehat{\pi}_{t+1} + \kappa\hat{y}_t + \varepsilon_t^S. \tag{1}$$

In this expression $\kappa = (\sigma + \varphi)(1-\theta)(1-\theta)/\theta$ is a combination of various underlying parameters of the economy: the elasticity of intertemporal substitution in consumption (σ), the labour elasticity (φ), the psychological discount factor (β) and the Calvo probability of price adjustment (θ). This relation comes from the aggregation of the supply decision of firms that have a market power and can re-optimize their selling price with discontinuities (*i.e.* as there are price rigidities, they cannot modify their selling price at any point in time). By so, they increase their price depending on four motives. The first motive is lagged inflation: this backward looking dynamics could be explained by firms' adaptive expectations in the price setting process. The second motive is anticipated inflation: as firms cannot re-optimize their price, they take into account future inflation to set their price today. The third term is the output gap: when firms set their price they take into account the difference between supply and demand, so that inflation reflects tensions on the goods market. Firms increase their prices during expansions while they decrease it during recessions. Finally, this relation incorporates a cost push term ε_t^S (such that: $\varepsilon_t^S > 0$ may indicate an increase in raw materials or energy price in the economy). We assume that ε_t^S is described by an iid AR(1) process such as:

$$\varepsilon_t^S = \rho^S\varepsilon_{t-1}^S + \eta_t^S, \tag{2}$$

with $\eta_t^S \sim \mathcal{N}(0, \sigma_S^2)$ where ρ^S denote respectively the AR term. The AR term aims at capturing some persistence in the diffusion of the shock while the MA term captures high fluctuations observed in inflation variations. The New Keynesian Phillips Curve (NKPC) is derived from the model of Calvo (1983), which combines staggered price-setting by imperfectly competitive firms. Specifically, Calvo assumes that each period a proportion θ of firms, randomly chosen, can reset their selling prices. Using this assumption, Clarida et al. (1999) show that the Phillips curve – the so-called New Keynesian Phillips Curve – then takes a particularly simple form in which inflation depends on the current gap between actual and equilibrium output as in the standard Phillips curve but on expected future inflation rather than on past inflation.

1.2.2 The IS curve

The second relation is the intertemporal (dynamic) IS curve. This schedule is a log linearization of the Euler bond equation that describes the intertemporal allocation of consumption of agents in the economy. Accounting for consumption habits (so as to introduce both forward and backward factors in consumption decisions), and using the equilibrium of the goods market to express the demand side of the economy in terms of the log deviation of activity, we get:

$$\hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma} \left(\hat{r}_t - \mathbb{E}_t \widehat{\pi}_{t+1} \right) + \varepsilon_t^D. \tag{3}$$

In this expression, σ is the elasticity of intertemporal substitution in consumption ($1/\sigma$ is here the elasticity of the output gap to the interest rate, i.e. the slope of the IS curve). This relation plays the same role as the IS curve in the IS-LM model. It is obtained from the intertemporal optimization of the welfare index of the agent subject to its budget constraint. Once log-linearized, this relation can be expressed in terms of the output gap. It links the current output gap ($\hat{y}_t$) to a linear combination of the real interest rate ($\hat{r}_t - \mathbb{E}_t \widehat{\pi}_{t+1}$) and of the expected ($\mathbb{E}_t \hat{y}_{t+1}$) output gap and to an exogenous preference shock ε_t^D (that will represent a demand shock). The demand shock is described by is an iid AR(1) process of the form:

$$\varepsilon_t^D = \rho^D \varepsilon_{t-1}^D + \eta_t^D, \tag{4}$$

with $\eta_t^D \sim \mathcal{N}(0, \sigma_D^2)$. This variable accounts for all the other features of the economy not taken into account in the structural part of model (mainly fiscal and international aspects).

1.2.3 The Monetary Policy Rule

The third relation is the Taylor rule that links the (log deviation of the) nominal interest rate $\hat{r}_t$ (controlled by monetary authorities) to the inflation rate ($\widehat{\pi}_t$) and to the output gap ($\hat{y}_t$). Monetary policy inertia is accounted for through the previous period rate of interest ($\hat{r}_{t-1}$). It is defined as:

$$\hat{r}_t = \rho \hat{r}_{t-1} + (1 - \rho) \left(\phi^\pi \widehat{\pi}_t + \phi^y \hat{y}_t \right) + \phi^{\Delta y} \Delta \hat{y}_t + \varepsilon_t^R.$$

In this expression, $\rho \in [0, 1)$ is the autocorrelation parameter, $\phi^\pi > 1$ is the elasticity of the nominal interest rate to the inflation rate, ϕ^y is the elasticity of the nominal interest rate to the output gap, $\phi^{\Delta y}$ is the elasticity to output growth and ε_t^R denotes a monetary policy shock that follows an *iid* AR(1) process such as:

$$\varepsilon_t^R = \rho^R \varepsilon_{t-1}^R + \eta_t^R, \tag{5}$$

with $\eta_t^R \sim \mathcal{N}(0, \sigma_R^2)$. This shock catches monetary policy decisions implying deviations from the standard Taylor rule like unconventional measures, or aimed at reshaping inflation expectations in the medium run. This relation is called a simple rule of monetary policy. It comes from studies in the conduct of monetary policy presented by Taylor (1993) and Clarida et al. (1999) that shows that most central banks follow this simple monetary rule.

1.3 Summary of the linear model and parametrisation

$$\text{AS curve}: \widehat{\pi}_t = \beta \mathbb{E}_t \widehat{\pi}_{t+1} + (\sigma + \varphi)(1 - \theta)(1 - \beta\theta)/\theta \hat{y}_t + \varepsilon_t^S. \tag{6}$$

$$\text{IS curve}: \hat{y}_t = \mathbb{E}_t \hat{y}_{t+1} - \frac{1}{\sigma} \left(\hat{r}_t - \mathbb{E}_t \widehat{\pi}_{t+1} \right) + \varepsilon_t^D. \tag{7}$$

$$\text{MP curve}: \hat{r}_t = \rho \hat{r}_{t-1} + (1 - \rho) \left(\phi^\pi \widehat{\pi}_t + \phi^y \hat{y}_t \right) + \phi^{\Delta y} \Delta \hat{y}_t + \varepsilon_t^R \tag{8}$$

$$\text{supply shock}: \varepsilon_t^S = \rho^S \varepsilon_{t-1}^S + \eta_t^S \tag{9}$$

$$\text{demand shock}: \varepsilon_t^D = \rho^D \varepsilon_{t-1}^D + \eta_t^D \tag{10}$$

$$\text{monetary policy shock}: \varepsilon_t^R = \rho^R \varepsilon_{t-1}^R + \eta_t^R \tag{11}$$

Description	Name	Value
Discount factor	β	0.99
Consumption Risk	σ	1
Labour disutility	φ	1
Calvo lottery	θ	0.75
Monetary policy smoothing	ρ	0.80
Inflation reaction	ϕ^{π}	1.5
Output gap reaction	ϕ^{y}	0.10
Output gap reaction	$\phi^{\Delta y}$	0.20
Supply shock AR(1) term	ρ^{S}	0.95
Demand shock AR(1) term	ρ^{D}	0.80
Monetary Policy shock AR(1) term	ρ^{R}	0.40
Supply shock standard deviation	η_t^{S}	0.40%
Demand shock standard deviation	η_t^{D}	2%
Monetary Policy shock standard deviation	η_t^{R}	0.1%

1.4 Impulse response function

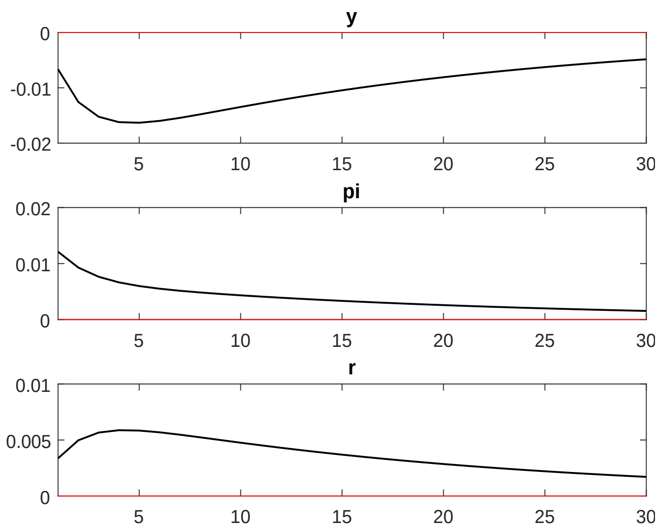


Figure 4: System response after a supply shock

Figure 4 reports the response of a new Keynesian economy after a supply shock characterized by an exogenous increase in the inflation rate. A positive inflation rate reduces real output by roughly 1.5% after 4 periods. To fight inflation, the central bank rise its interest rate by 6 bpts which in turn deteriorates output through the IS curve and reduces inflation. The effect of supply shocks are rather persistent, here the economy requires up to 40 quarters before getting back to equilibrium.

The second shock reported in Figure 5 is a demand shock characterized by an exogenous increase in the output gap, reflecting a gap between the demand and the supply on the goods market. The market clears through an increase in prices, which in turn forces the central bank to react by a monetary policy tightening.Finally, the last shock is a monetary policy shock reported in Figure 6. This shock is characterized by an exogenous increase in the policy rate, *i.e.* a particular situation where the central bank decides to leave temporary its own Taylor rule (*e.g.* inflation expectations reshaping, unconventional measures, zero lower bound constraint, etc). A monetary policy tightening reduces the aggregate output

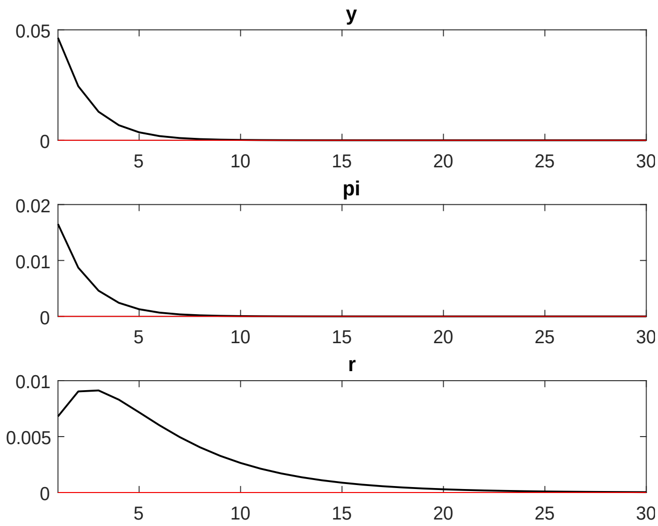


Figure 5: System response after a demand shock

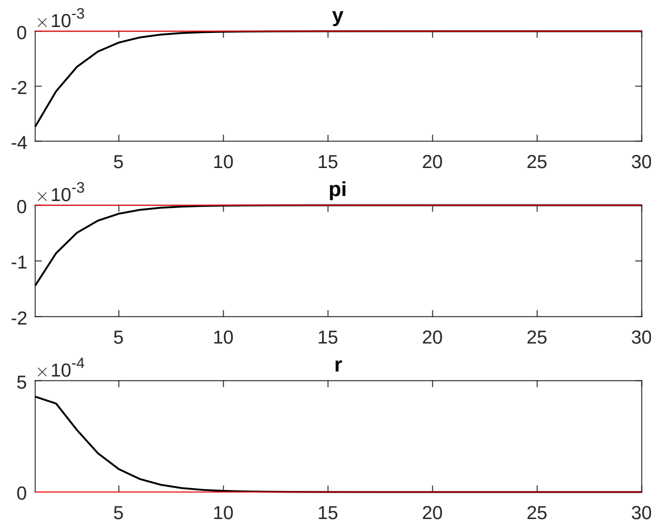


Figure 6: System response after a monetary policy shock

through the IS curve, which in turn generate a disequilibrium on the goods market characterized by an excess of goods supply cleared by a decrease in the inflation rate.

Overall, our framework is also consistent with the BVAR model as systems responses between models are quite similar. To compare the BVAR with the DSGE on a regular basis, we should have plotted the growth of the GDP rather than output gap.

2. Bayesian econometrics

In this section, the methodology regarding mapping the model to the data is discussed as well as the priors setting.

2.1 Methodology

We estimate the 3-equation model on US data using Bayesian techniques as in Fernandez-Villaverde and Rubio-Ramirez (2004) and Smets and Wouters (2007). Non-calibrated parameters are estimated using the Bayesian maximum likelihood principle (see An and Schorfheide (2007) for an extensive presentation). As exemplified by Smets and Wouters (2003) and Smets and Wouters (2007), this approach can fruitfully be applied to macroeconomics as it allows a DSGE model to perform as well as non-constrained VAR model in explaining macroeconomic time series. In our estimation we follow closely each of the steps defined by Smets and Wouters (2003).[2] As defined in the model, the aim is to estimate the vector of parameters Θ:

$$\Theta = \left[\begin{array}{cccccccccccccc} \sigma & \varphi & \phi^{\pi} & \phi^{y} & \phi^{\Delta y} & \rho & \theta & \sigma_S & \sigma_D & \sigma_R & \rho^S & \rho^D & \rho^R & \end{array} \right]', \tag{12}$$

while the discount factor β is rather calibrated as this parameter is known not to be well identified.

The Bayesian approach consists in defining an *apriori* distribution $p(\Theta)$ on the structural parameters of Θ that concerns both a prior value for this parameter and a distribution. The Bayesian estimate of a parameter thus mixes the prior mean with the maximum likelihood estimate. The choice of the initial values for the prior parameters requires an intuition for their role in the economy. We follow the standard practice of the literature in choosing one of the following probability distribution for parameters: the Normal distribution ($\mathcal{N}$), the Beta distribution ($\mathcal{B}$, for parameters with values lying between 0 and 1), the Gamma distribution ($\mathcal{G}$, for priors which allow relatively diffuse and positive support) and the Inverse Gamma distribution ($\mathcal{IG}$, for shocks, as this distribution allows us to choose priors that are strongly diffuse and positive). The estimates of the parameters are updated using the dataset Z according to Bayes' rule:

$$p(\Theta|Z) = \frac{p(\Theta|Z)p(\Theta)}{p(Z)} \approx L(\Theta|Z)p(\Theta), \tag{13}$$

where $L(\Theta|Z) = p(Z|\Theta)$ denote the likelihood function, $p(\Theta|Z)$ is the posterior distribution of parameters and $p(Z)$ is the marginal likelihood (marginal data density). Our model thus forms a linear system with rational expectations and the solution is given by:

$$X_t = M_1(\Theta)X_{t-1} + M_2(\Theta)\Sigma_t, \tag{14}$$

$$\Sigma_t = M_3(\Theta)\Sigma_{t-1} + M_4(\Theta)\Gamma_t, \tag{15}$$

where $X_t = \left[\begin{array}{ccc} \Delta\hat{y}_t & \hat{\pi}_t & \hat{r}_t \end{array} \right]'$ is a vector of endogenous variables, $\Sigma_t = \left[\begin{array}{ccc} \varepsilon_t^D & \varepsilon_t^S & \varepsilon_t^R \end{array} \right]'$, $\Gamma_t = \left[\begin{array}{ccc} \eta_t^D & \eta_t^S & \eta_t^R \end{array} \right]'$ and $M_i(\Theta)$, for $i = 1, 2, 3, 4$, are matrices depending on the parameters of the model. Finally, measurement equations, that link observable variables Z_t used in the estimation to the endogenous variables of the model X_t, are defined as:

$$Z_t = \bar{X} + X_t, \tag{16}$$

[2] The successive steps of this estimation strategy is presented by Smets and Wouters (2003) as follows: "this econometric approach attempts to provide a full characterisation of the observed data series. For example, following Sargent (1989), a number of authors have estimated the structural parameters of DSGE models using classical maximum likelihood methods. These maximum likelihood methods usually consist of four steps. In the first step, the linear rational expectations model is solved for the reduced form state equation in its predetermined variables. In the second step, the model is written in its state space form. This involves augmenting the state equation in the predetermined variables with an observation equation which links the predetermined state variables to observable variables. In this step, the researcher also needs to take a stand on the form of the measurement error that enters the observation equations. The third step consists of using the Kalman filter to form the likelihood function. In the final step, the parameters are estimated by maximising the likelihood function. Alternatively within this strong interpretation, a Bayesian approach can be followed by combining the likelihood function with prior distributions for the parameters of the model, to form the posterior density function. This posterior can then be optimised with respect to the model parameters either directly or through Monte-Carlo Markov-Chain (MCMC) sampling methods."

Shape	Name	Support	Example(s)
normal_pdf	$\mathcal{N}(\mu, \sigma)$	$(-\infty, +\infty)$	Policy parameters, utility curvatures,
gamma_pdf	$\mathcal{G}(\mu, \sigma)$	$(0, +\infty)$	Elasticities, utility curvatures, trends
beta_pdf	$\mathcal{B}(\mu, \sigma)$	$[0, 1]$	Probabilities, AR-MA terms, shares...
inv_gamma_pdf	$\mathcal{IG}_1(\mu, \sigma)$	$(0, +\infty)$	Standard deviations
uniform_pdf	$\mathcal{U}(\mu, \sigma)$	$(-\infty, +\infty)$	Same as normal distribution but with lower/upper bounds
weibull_pdf	$\mathcal{W}(\mu, \sigma)$	$[0, +\infty)$	For standard deviations, in particular of measurement shocks
inv_gamma2_pdf	$\mathcal{IG}_2(\mu, \sigma)$	$(0, +\infty)$	Standard deviations

Table 1: Prior distributions available in Dynare

where $\bar{X}$ denotes the vector of the estimated mean of endogenous variables. Here, $\bar{X}$ is vector of all zeros as for tractability we don't estimate trend components.

Equations (14)-(16) form the state-space representation of the model, the likelihood of which can be evaluated using the Kalman filter. Obtaining the analytical expression for the likelihood function is generally not possible. However, the posterior distribution of the model parameters can be constructed numerically by applying the Markov Chain Monte Carlo (MCMC) methods. Finally, we compute the posterior moments of the parameters using a sufficiently large number of draws (here $50,000$) for 4 different chains, having made sure afterwards the MCMC algorithm has converged.

One main advantage of DSGE models compared to VARs concerns the evaluation of monetary policy. DSGE models replicate time series by the responses of agents to different sequences of shocks. Once we have estimated all the sequences of shocks and the reaction parameters of households, firms and monetary authorities, we can perform counterfactual analyses by changing the parameters of monetary policy. Counterfactual scenarios can be derived from the estimated model by optimizing the policy coefficients in the Taylor to see how well the economy would have performed under an alternative policy regime.

2.2 Priors

When we are estimating with Bayesian techniques, we need to specify prior distributions for each parameter to be estimated. In the estimated params section, for each estimated parameter, we declare a distribution, and specify parameters relevant for that distribution. The syntax for this is:

[parameter_name], [initial_value(optional)], [lower_bound(optional)], [upper_bound(optional)], [prior_shape], [prior_mean] , [prior_standard_deviation], [p_3(optional)], [p_4(optional)], [jscale(optional)] ;

Table 1 details the distributions. Prior specification (in terms of shape, mean and standard deviation) allows to give more information about the distribution of the parameters and to avoid some regions where the likelihood function could be flat.

To illustrate the guideline for choosing a prior, let us consider first the AR(1) term of the shock processes ρ^S, ρ^D and ρ^R. To have a stationary model, these parameters must be below 1 while they must be above 0 get a smooth response. The beta distribution meets this requirement as it has a support on $[0, 1]$ and will not allow this parameter above 1 or below 0. In estimated_params; block, we can add the following 3 parameters:

```
estimated_params;
        stderr eta_D,          , ,
         ↪   ,        INV_GAMMA_PDF,              .1,        2;
        stderr eta_S,            , ,
         ↪   ,        INV_GAMMA_PDF,              .1,        2;
```

```
        stderr eta_R,        , , ,        INV_GAMMA_PDF,                .1,        2;
end;
```

In the previous specification, we assume that each parameter is likely to have a mean 0.5 and a standard deviation of 0.20. This prior is quite uninformative as the support of this distribution allows parameters to be estimated near 1 or near 0. Conversely, an opinionated prior with a mean of 0.90 and a standard deviation of 0.01 would allow the parameter to be estimate very close to 0.90. Economists are free to calibrate their prior, and in turn give or not information about the estimation of the parameter. For instance here, we know that monetary policy shocks are not strongly autocorrelated, a prior information of 0.20 with standard deviation of 0.05 could be fairly acceptable. Even through uninformative priors are preferred, it is possible to set tight priors but they have to be carefully justified.

Let us consider another example. The estimated standard deviation of a shock has to be positive, this constraint excludes *de facto* the Normal distribution as it has a negative support. A Beta distribution could be justified as we want estimated shocks to be small otherwise our first order approximation becomes irrelevant with large shocks. However, demand and supply shocks may have a standard deviation higher than 1, which excludes the Beta distribution. It is then possible to use a Uniform distribution, as well as a Gamma or Inverse Gamma distributions. Here, we follow Smets and Wouters (2007) using an Inverse Gamma distribution with a mean of 0.10 and a standard deviation of 2. Accordingly, we add in the Dynare file the lines which code the estimation of the shocks' volatility.

```
        stderr eta_D,           , ,
         ↪  ,        INV_GAMMA_PDF,               .1,        2;
        stderr eta_S,             , ,
         ↪  ,        INV_GAMMA_PDF,               .1,        2;
        stderr eta_R,        , , ,        INV_GAMMA_PDF,                .1,        2;
```

2.3 Estimation and posteriors

Once the priors are written in Dynare (which I have not totally detailed), we have to declare which variables are observable (and they should appear in the data file as well). Here, we estimate using our three time series:

```
varobs         y_obs pi_obs r_obs;
```

Finally, we declare the estimation command as we did for maximum likelihood estimations. There are lots of options for Bayesian estimation which can go here; it's worth seeing the Dynare Manual for an overview of these options, and resources on Bayesian estimation for what they mean.

```
estimation(datafile=mydata,first_obs=1,mode_compute=4,
mh_replic=20000,mh_jscale=0.5,prefilter=1,lik_init=2,bayesian_irf,irf=30) y_obs pi_obs
 ↪  r_obs;
```

Here, mh_replic denotes the number of draws sampled by Metropolis-Hastings algorithm which aims at drawing the posterior distributions of estimated parameters. Increasing the number of draws allows to obtain converging chains, nonetheless it is not a sufficient condition for convergence. For small models, the number of draws can be low (roughly here 50 000 draws is fine), for large scale model, it is preferable to have a larger number of draws to get converging chains (for instance, Smets and Wouters (2007) draws 200 000 draws). Option lik_init set to 2 allows Dynare to handle non-stationnary observable variables in the fit exercise, we know that the GDP is a non-stationary variable requiring a special treatment for the estimation. Option prefilter remove the empirical mean in the observable matrix (i.e. it demeans our sample). Estimating theoretical means would require to had a trend component in the production function, which we don't do here. Finally, option mh_jscale is a parameters which has to be adjusted in order to obtain an acceptance ratio between 20% and 30% (if the final acceptance rate is below 20%, we have to decrease mh_jscale while if it is below we have to increase 30%). The interval for the acceptance ratio relies on research on bayesian statistics which I won't develop here.

Starting the estimation exercise, Dynare loads our data file, and starts to find the posterior mode using csminwel algorithm using optimizations methods. The initial value of the log-likelihood function is -9295.463, the algorithm employed by Dynare aims at maximising the value of the log-likelihood function by adjusting the estimated parameters.

console response

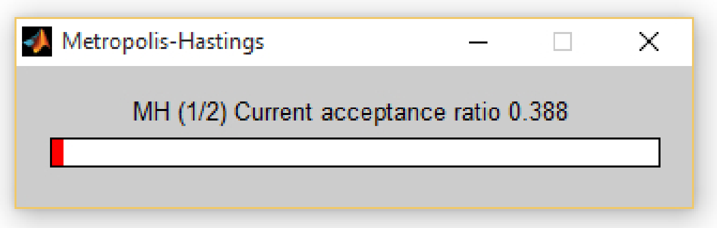
```
Initial value of the log posterior (or likelihood): -9295.463
____________
f at the beginning of new iteration, 9295.4630399514
Predicted improvement: 2095725.132776231
lambda = 1; f = 9405.1065976
lambda = 0.33333; f = 9305.6233091
lambda = 0.11111; f = 9296.0459801
lambda = 0.037037; f = 9295.4666624
lambda = 0.012346; f = 859.3227388
lambda = 0.0041152; f = 2326.0104571
Norm of dx 20.473
```

At the final iteration of the algorithm (here 58 iterations), the improvement performed in the last iteration is low enough for the algorithm to stop the optimisation. The value of the log-likelihood function is now -318.102525. Dynare then prints in the console the mode of the posterior distribution:

console response

```
RESULTS FROM POSTERIOR ESTIMATION
parameters
 prior mean mode s.d. prior pstdev
rho_D 0.500 0.9476 0.0153 beta 0.2000
rho_S 0.500 0.9902 0.0068 beta 0.2000
rho_R 0.500 0.1494 0.0610 beta 0.2000
sigma 1.000 2.5692 0.4385 gamm 0.4000
varphi 2.000 1.2936 0.5457 gamm 0.7500
phi_pi 1.500 1.6380 0.0798 norm 0.1000
phi_y 0.050 0.0363 0.0180 norm 0.0500
phi_dy 0.050 0.1747 0.0252 norm 0.0500
rho 0.750 0.7739 0.0256 beta 0.0500
theta 0.500 0.8780 0.0220 beta 0.1000
standard deviation of shocks
prior mean mode s.d. prior pstdev
eta_D 0.100 0.0672 0.0129 invg 2.0000
eta_S 0.100 0.0649 0.0202 invg 2.0000
eta_R 0.100 0.2306 0.0136 invg 2.0000
```

After finding the mode, the Metropolis-Hastings algorithm is employed. For each chain, it samples the number of draws we specified in the estimation() command:



Here, we have two chains of 20 000 draws. At the end of the sampling, we will get 40 000 samples. Dynare will automatically drop a certain percentages of draws (the percentage of drops can be manually set in the estimation command). Dynare compute the uncertainty (confidence intervals size can be manually set in the estimation command) and the mean of each parameter using the sequence of random samples from Metropolis-Hastings output.

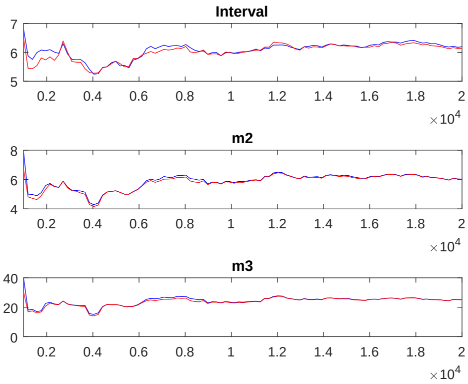


Figure 7: Univariate convergence diagnostic, Brooks and Gelman (1998)

console response

```
parameters
 prior mean post. mean 90% HPD interval prior pstdev
rho_D 0.500 0.9459 0.9203 0.9690 beta 0.2000
rho_S 0.500 0.9890 0.9805 0.9985 beta 0.2000
rho_R 0.500 0.1657 0.0678 0.2573 beta 0.2000
sigma 1.000 2.5887 1.8232 3.2544 gamm 0.4000
varphi 2.000 1.5169 0.5959 2.4169 gamm 0.7500
phi_pi 1.500 1.6645 1.5350 1.7889 norm 0.1000
phi_y 0.050 0.0333 0.0069 0.0591 norm 0.0500
phi_dy 0.050 0.1734 0.1361 0.2140 norm 0.0500
rho 0.750 0.7625 0.7172 0.8056 beta 0.0500
theta 0.500 0.8717 0.8365 0.9066 beta 0.1000
standard deviation of shocks
 prior mean post. mean 90% HPD interval prior pstdev
eta_D 0.100 0.0696 0.0489 0.0897 invg 2.0000
eta_S 0.100 0.0767 0.0419 0.1134 invg 2.0000
eta_R 0.100 0.2367 0.2149 0.2609 invg 2.0000
```

Dynare prints the mean and 90% confidence interval upper and lower bounds as well as the prior standard deviation of the parameter. The posterior mean is preferred than the mode for empirical exercises. Only one parameter is not statistically significant, the policy stance on output ϕ^y as its confidence interval includes 0 at 90%. Dynare also runs some diagnostics to assess the convergence.

Figure 7 reports some of the diagnostics computed by Dynare to examine the convergence of the MCMC chains as in Brooks and Gelman (1998). Here for instance, the red and blue lines should converge otherwise we should increase the number of draws, or adjust the priors/model.

Figure 8 reports the prior and posterior distributions of parameters and shocks involved in the fit exercise. It is preferable to have a posterior distribution different than the prior one. Having the posterior diverge from the prior means that your data (the likelihood in Bayes rule) is very informative about the parameter (if the posterior distribution is not too wide around the mode). If all your posterior estimates would coincide with the prior mean, there would be no reason for estimation as the data does not add any new insights. In this situation, you should calibrate rather than estimate when the identification of your

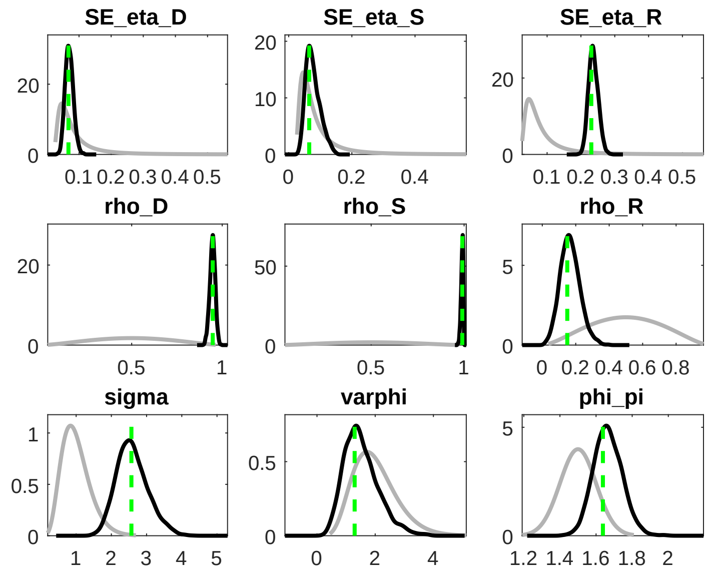


Figure 8: Priors (grey) and posteriors (black) distributions of estimated parameters and mean (green).

parameter is bad.

For instance in Figure 8, ρ^D and ρ^S are clearly identified, i.e. data were informative for these parameters as their priors and posteriors are different. Conversely, data were not informative for the identification of the labour utility curvature φ. The identification of parameters in the utility function is known to be challenging, these parameters are generally calibrated.

In our situation, this model is not acceptable for a publication for 3 reasons:

1. The identification of φ is bad.
2. ρ^D and ρ^S captures too much information and their estimated means are too high, revealing a potential unit root problem the current set-up cannot handle (need to include a trend which some stochastic shock on it).
3. The acceptance ratio is roughly 40%, we should increase the mh acceptance rate.

Stored in	Description
oo_.posterior_mode	Parameters and shock standard errors at the posterior mode.
oo_.posterior_std	Standard error of the estimated values at the posterior mode.
oo_.posterior_density	Density of the posterior for each parameter (x,y values for each parameter).
oo_.prior_density	Density of the prior for each parameter (x,y values for each parameter).
oo_.SmoothedVariables	Estimated values of each endogenous variable for all time periods.
oo_.SmoothedShocks	Estimated values of each shock for all time periods.

Table 2: Matlab's workspace after a Bayesian estimation

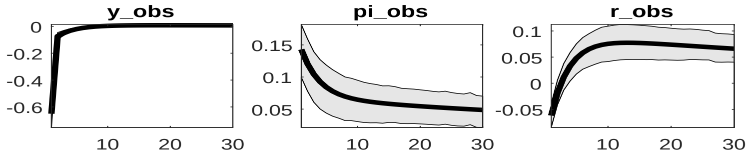


Figure 9: System response after an estimated supply shock

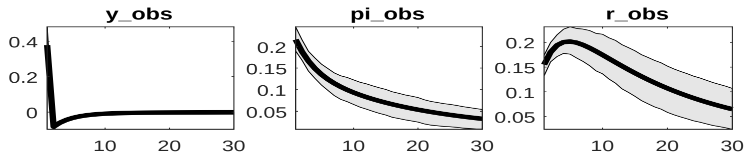


Figure 10: System response after an estimated demand shock

2.4 Bayesian system response

It is also worthwhile to have a look at the bayesian system response which can easily be compared to the IRF of the BVAR(4) (a conflict with my Matlab version seems to have broken the plot of confidence intervals).

Figure 9, Figure 10 and Figure 11 report the system response of the model to an estimated supply, demand and monetary policy shock (again, confidence intervals are broken). The high (too) auto-correlation of the AR(1) shock process generates a non-stationary response of output in case of a supply shock. Overall, the general response of the model is consistent with the results of the VAR.

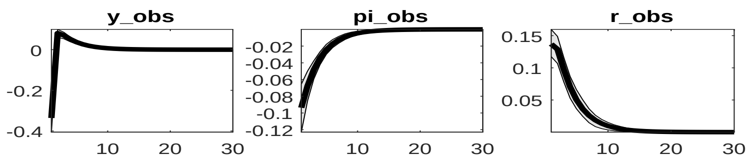


Figure 11: System response after an estimated monetary policy shock

Exercise 1: Empirical applications

First, it is possible to load back the results instead of doing the estimation each time you run the model. Once your model has been estimated, replace the estimation function by:

console response

```
estimation(datafile=mydata,first_obs=1,mh_replic=0,mh_jscale=0.5,prefilter=1,lik_init=2,load_mh
y y_obs pi_obs r_obs;
```

Using the estimated DSGE model, answer the following questions:

1. Plot the shock decomposition. To do so, after the estimation, use the command: shock_decomposition y_obs pi_obs r_obs;
   What are the drivers of the business cycles?

2. What are the driving forces of each observable up to 1, 5, 10, 100 quarters? (after estimation command, use the stoch_simul function with option conditional_variance_decomposition=[1,5,10,100]).

3. Is the model able to replicate the empirical standard deviation of the observable variables?

4. Using option forecast, perform a prediction exercise from 1 to 8 quarters. Use Dynare User Guide to find the good syntax.

Exercise 2: Fixing the model

Recall that in our situation, this model is not acceptable for a publication for 3 reasons: (i) the identification of φ is bad, (ii) parameters ρ^D and ρ^S captures too much information and their estimated means are too high, revealing a potential unit root problem the current set-up cannot handle; (iii) the acceptance ratio is roughly 40%, we should increase the mh acceptance rate. Given these 3 failures:

1. Fix the identification problem, calibrating the parameter or adjusting its prior information (try a beta distribution for instance).

2. To fix the information capture by AR(1) terms, our model needs to be more backward looking. Add consumption habits, denoted h, and inflation indexation ξ in the relevant equations. The following backward looking equations are:

$$\begin{aligned}\widehat{\pi}_t &= \frac{\xi}{1+\xi\beta}\widehat{\pi}_{t-1} + \frac{\beta}{1+\xi\beta}\mathbb{E}_t\widehat{\pi}_{t+1} + \kappa\hat{y}_t + \varepsilon_t^S,\\ \hat{y}_t &= \frac{h}{1+h}\hat{y}_{t-1} + \frac{1}{1+h}\mathbb{E}_t\hat{y}_{t+1} - \frac{1}{\sigma\left(1+h\right)}\left(\hat{r}_t - \mathbb{E}_t\widehat{\pi}_{t+1}\right) + \varepsilon_t^D.\end{aligned}$$

   and estimate these parameters. Is the model fit improved? Does it fix the excessive autocorrelation?

3. Adjust the acceptance rate.

4. Evaluate the empirical relevance of an ARMA(1,1) supply shock as in Smets and Wouters (2007):

$$\varepsilon_t^S = \rho^S\varepsilon_{t-1}^S + \eta_t^S - u\eta_{t-1}^S$$

   where u is the MA term.

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Appendices

A. Micro-foundations of the 3-equation model

A.1 Households

There is a continuum of households $j \in [0;1]$ with a utility function

$$U\left(C_{jt}, H_{jt}\right) = \left[\frac{C_{jt}^{1-\sigma}}{(1-\sigma)} - \chi \frac{H_{jt}^{1+\varphi}}{(1+\varphi)}\right].$$

The utility is assumed to increase in consumption and to decrease with labour. The representative household maximizes its welfare, defined as the expected stream of utilities discounted by $\beta \in (0,1)$:

$$\max_{C_{jt}, H_{jt}, B_{jt}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^{\tau} U\left(C_{jt+\tau}, C_{t-1+\tau}, H_{jt+\tau}\right) \tag{17}$$

Under the budget constraint:

$$P_t C_{jt} + e^{\delta \varepsilon_t^D} B_{jt+1} = R_{t-1} B_{jt} + W_t H_{jt}, \tag{18}$$

where $\sigma > 0$ and $\varphi > 0$ are shape parameters of the utility function with respect to consumption C_{jt} and to hours worked $H_{jt+\tau}$ while χ is the a shift parameter which scales the steady state labour supply to realistic values. Prices, denoted P_t, is an aggregate of price of all varieties of goods produced in the economy. Following the microeconomic evidence, households are assumed to have deep habits in their consumption plans.

Households save each period an amount B_{jt+1} in t and is remunerated the next period B_{jt} at a riskless rate R_{t-1} decided by the central bank. For stability purpose of the model in a rational expectations equilibrium, the interest rate is decided at the very end of $t-1$. As Smets and Wouters (2007), we introduce an AR(1) demand shock process in the budget constraint of the representative household denoted ε_t^D multiplied by δ in order to normalize to unity the shock in the linear form of the Euler equation. Finally, the representative household is involved in the production process of goods through its labour supply H_{jt} to firms, in return the household obtain a nominal wage W_t.

Consumption C_{jt} is a composite of different varieties i of goods produced in the economy defined by, $C_{jt} = (\int_0^1 C_{jit}^{(\epsilon-1)/\epsilon} \mathrm{d}i)^{\epsilon/(\epsilon-1)}$, with substitutability degree u_t. The corresponding aggregate price of all varieties in the economy is determined by, $P_t = (\int_0^1 P_{it}^{1-\epsilon} \mathrm{d}i)^{1/(1-\epsilon)}$. The associated demand for each variety by the representative household is, $C_{jit} = (P_{it}/P_t)^{-\epsilon} C_{jt}$.

The household j face the budget constraint over time, the dynamic Lagrangian problem thus writes:

$$\mathcal{L} = \mathbb{E}_t \sum_{\tau=0}^{+\infty} \beta^{\tau} \left(\begin{array}{c} \left[\frac{C_{jt}^{1-\sigma}}{(1-\sigma)} - \chi \frac{H_{jt}^{1+\varphi}}{(1+\varphi)}\right] \\ -\lambda_{t+\tau}^c \left[P_{t+\tau} C_{jt+\tau} + e^{\delta \varepsilon_{t+\tau}^D} B_{jt+1+\tau} - R_{t-1+\tau} B_{jt+\tau} - W_{t+\tau} H_{jt+\tau}\right] \end{array} \right)$$

where λ_t^c denotes the Lagrange multiplier associated to the budget constraint. This lagrange multiplier also denotes the marginal utility of consumption or the shadow value of one unit of consumption. This lagrange multiplier is important to determine how the economy discounts the future. The discounting in t of any variable in $t+1$ is achieved through the stochastic discount factor denoted, $\beta \lambda_{t+1}^c / \lambda_t^c$.[3] First order

[3] In equilibrium, the stochastic discount factor is determined by the central bank through the Euler equation.

conditions which maximizes the welfare index under the budget constraint read as follows:

$$(C_{jt}) : C_{jt}^{-\sigma} - P_t \lambda_t^c = 0$$
$$(B_{jt+1}) : -\lambda_t^c e^{\delta \varepsilon_t^D} + \beta \lambda_{t+1}^c R_{t-1} = 0$$
$$(H_{jt}) : -\chi H_{jt}^{\varphi} + \lambda_t^c W_t = 0$$

Substituting the Lagrange multiplier λ_t^c, we find the first order conditions that solve the optimization problem. First, we find Euler bond condition which determines the utility maximizing consumption path C_{jt}:

$$\left(\frac{C_{jt}}{C_{jt+1}}\right)^{-\sigma} = \frac{\beta}{e^{\delta \varepsilon_t^D}} \mathbb{E}_t \frac{R_t}{\pi_{t+1}}, \tag{19}$$

where $\pi_{t+1} = P_{t+1}/P_t$ is the expected inflation rate, now parameter σ also denotes elasticity of the consumption growth to any expected change of the real interest rate $\mathbb{E}_t R_t/\pi_{t+1}$. The demand shock ε_t^D, deflects temporary the household from its optimal consumption path and forces her to spends more in case of a positive demand shock.

Second, we obtain the labour supply equation, this equation implies that at the optimal rate of substitution between labour and consumption equals the real wage W_t/P_t. This equation reads as follows:

$$\chi C_{jt}^{\sigma} H_{jt}^{\varphi} = \frac{W_t}{P_t} \tag{20}$$

A.2 Firms

The producing sector is composed of two types of complementary firms, namely final and intermediates firms, which produced goods in the economy.

A.2.1 Final firms

The final good producers are retailers, they buy intermediate goods Y_{it}, package them into Y_t^d, and sell the final good to households. In equilibrium on the good market, Y_t^d also denotes the aggregate demand for goods from households. On a perfectly competitive market, final producers maximize profits:

$$P_t Y_t^d - \textstyle\int_0^1 P_{it} Y_{it} \mathrm{d}i,$$

subject to a supply constraint:

$$Y_t^d = \left(\textstyle\int_0^1 Y_{it}{}^{(\epsilon-1)/\epsilon} \mathrm{d}i\right)^{\epsilon/(\epsilon-1)}. \tag{21}$$

This supply constraint implies that final producers have a technology which aggregate non-perfectly substitutable goods. This imperfect substitutability between all types of varieties i is driven by the monopolistic competition on the intermediate good market. Each good i is an imperfect substitute, allowing intermediate firms to gain positive profits through a gap between their selling and producing price. Goods are not perfect subsititutes, $\epsilon > 1$ denotes the imperfect substitutability between different goods varieties allowing firms to set prices above their marginal cost which induces positive profits and monopolistic competition. The intensity of the monopolistic competition is driven by $\epsilon/(\epsilon - 1)$ the mark-up over the producing price of intermediate firms.

From a microeconomic perspective, the supply shock captures changes in the margins of firms over the business cycles. From a macroeconomic perspective, this supply shock catches up shocks to the inflation equation, such as oil price shock, commodity prices shock, etc.

Finally, we find the intermediate demand function for each variety i is:

$$Y_{it} = \left(\frac{P_{it}}{P_t}\right)^{-\epsilon} Y_t^d, \ \ \forall i \tag{22}$$

A.2.2 Intermediate firm

Representative intermediate good producer i uses labor inputs to produce a variety i of good on a monopolistically competitive market. Each firm is specialized in the production of one variety. To introduce nominal rigidities, we decompose the production process of goods in two steps, this implies the firm solves a two-stage problem. In a first stage, the firm minimizes its production cost on a perfectly competitive input factors market.

The supply of goods for each firm is achieved by the following Cobb-Douglas production technology:

$$Y_{it} = H_{it}. \tag{23}$$

where Y_{it} is the production of variety i and H_{it} is the labour demand. The profit optimization problem of the representative firm thus writes:

$$\max_{H_{it},Y_{it}} MC_{it}Y_{it} - W_t H_{it} \tag{24}$$

where MC_{it} denotes the marginal cost of one unit of good. The (static) Lagrangian problem thus write:

$$\mathcal{L} = MC_{it}Y_{it} - W_t H_{it} - \lambda_t \left[Y_{it} - H_{it}\right]$$

where λ_t denotes the lagrange multiplier on the supply constraint. The first order condition which minimizes the input costs write:

$$(H_{it}) : -W_t + \lambda_t = 0 \tag{25}$$

$$(Y_{it}) : MC_{it} - \lambda_t \tag{26}$$

In equilibrium, the marginal cost equals the lagrange multiplier. The marginal cost of producing one good is:

$$MC_{it} = W_t \tag{27}$$

In the second stage problem, the firms decides its selling price on a Calvo basis with nominal rigidities. This nominal rigidity prevent firms to optimally set prices. There is a fraction of firms θ that is not allowed to reset price, prices then evolves according to $P_{it} = P_{it-1}$ while for the other remain share of firms $1-\theta$, they are able to set their selling price such that $P_{it} = P^*_{it}$, where P^*_{it} denote the optimal price set by the representative firm.

$$P_{it} = \begin{cases} P^*_{it} \text{ with probability } 1-\theta \\ P_{it-1} \text{ with probability } \theta \end{cases}$$

Due to the nominal rigidity, the price sets now by the firm may last more than one period, which will affect the expected profits of the firms. Firms allowed to choose their new price P^*_{it} will consider the perspective of being price constrained in the future. The prix chosen persists over time according to:

$$\begin{array}{llllllll}
P^*_{it} & \rightarrow & (1-\theta)\,P^*_{it+1} & & & & & \\
 & \rightarrow & \theta P^*_{it} & \rightarrow & (1-\theta)\,P^*_{it+2} & & & \\
 & & & & \theta P^*_{it} & \rightarrow & (1-\theta)\,P^*_{it+3} & \\
 & & & & & \rightarrow & \theta P^*_{it} & \rightarrow \ldots
\end{array}$$

Summarizing the previous table, the firm allowed to reset its selling price with a probability $1-\theta$ maximizes the following expected sum of discounted profits:

$$\max_{P^*_{it}} \mathbb{E}_t \sum_{\tau=0}^{+\infty} \frac{\lambda^c_{t+\tau}}{\lambda^c_t} (\beta\theta)^\tau \left[\frac{P^*_{it}}{P_{t+\tau}} - \frac{MC_{it+\tau}}{P_{t+\tau}}\right] Y_{it+\tau} \tag{28}$$

Since firms are owned by households, they discount the expected profits using the same discount factor than households ($\beta^{\tau}\lambda^c_{t+\tau}/\lambda^c_t$). The firm face the downward sloping constraint from final good producers (re-arranged because of nominal rigidities):

$$Y_{it+\tau} = \left(\frac{P^*_{it}}{P_{t+\tau}}\right)^{-\epsilon} Y_{t+\tau}, \quad \tau > 0$$

The first order condition, which defines the optimal price setting given the probability θ to be price constrained in the future, is determined by:

$$\mathbb{E}_t \sum_{\tau=0}^{+\infty} \frac{\lambda^c_{t+\tau}}{\lambda^c_t} \left(\beta\theta\right)^{\tau} \left[\frac{P^*_{it}}{P_{t+\tau}} - e^{\varkappa\varepsilon^S_t} \frac{\epsilon}{\epsilon-1} \frac{MC_{it+\tau}}{P_{t+\tau}}\right] Y_{it+\tau} = 0.$$

As in Smets and Wouters (2003), the markup $\exp(\varkappa\varepsilon^S_t)\epsilon/\left(\epsilon-1\right)$ is assumed to be exogenously varying over time, ϵ denotes the imperfect substitutability between different goods varieties, ε^S_t denotes the supply shock and $\varkappa$ is a scale parameter that normalizes the shock to unity in the log-linear form of the model as in Smets and Wouters (2007).

A.3 Authorities

To close the model, the monetary policy authority sets its interest rate according to a standard Taylor Rule:

$$\frac{R_t}{\overline{R}} = \left(\frac{R_t}{\overline{R}}\right)^{\rho} \left(\left(\frac{\pi_t}{\overline{\pi}}\right)^{\phi^{\pi}} \left(\frac{Y_t}{\overline{Y}}\right)^{\phi^{y}}\right)^{1-\rho} \left(\frac{Y_t}{Y_{t-1}}\right)^{\phi^{\Delta y}} e^{\varepsilon^R_t},$$

where R_t is the nominal interest rate, π_t is the inflation rate, Y_t is the level of output and ε^R_t is an AR(1) monetary policy shock. Finally, parameters $\overline{R}$, $\overline{\pi}$ and $\overline{Y}$ are long term values for the interest rate, the inflation rate and the GDP.[4] The central bank reacts to the deviation of the inflation rate and the GDP from their steady state values in a proportion ϕ^{π} and ϕ^{y}, the central bank also smooths her rate at a degree ρ.

A.4 Equilibrium conditions

The resource constraint for the economy is defined by the aggregate demand from households, $Y^d_t = \int_0^1 C_{jt}\mathrm{d}j$. Aggregating the demand from final goods producers in Equation 22 is, $\int_0^1 Y_{it}\mathrm{d}i = \int_0^1 \left(P_{it}/P_t\right)^{-\epsilon}\mathrm{d}i$ Y^d_t . Letting $Y_t = \int_0^1 Y_{it}\mathrm{d}i$ and $C_t = \int_0^1 C_{jt}\mathrm{d}j$ respectively denote the aggregate supply and private consumption, the goods market equilibrium is then determined by:

$$Y_t = C_t \tag{29}$$

The labour market equilibrium between demand from firms and supply from households simply reads:

$$H_t = \int_0^1 H_{jt}\mathrm{d}j = \int_0^1 H_{it}\mathrm{d}i \tag{30}$$

Finally, the aggregation between constrained firms and non-constrained firms leads to the following equation for the aggregate prices:

$$\left(P_t\right)^{1-\epsilon} = \theta\left(P_{t-1}\right)^{1-\epsilon} + (1-\theta)\left(P^*_t\right)^{1-\epsilon} \tag{31}$$

[4] With a credible central bank, $\overline{\pi}$ and $\overline{Y}$ can also be interpreted as the targets of the central bank in terms of inflation rate and GDP.

A.5 Steady State and linearization

First we normalize prices $\bar{P}=1$ while we assume that households work one third of their time $\bar{H}=1/3$. Then we find:

$$\overline{C}=\overline{Y}=\overline{H},\overline{W}=\overline{MC}=\frac{\epsilon-1}{\epsilon},\chi=\overline{WC}^{-\sigma}\overline{H}^{-\varphi} \tag{32}$$

We linearized the relations around this steady state. First, combining the Euler bond equation and the resources constraint, *i.e.* $\hat{y}_t=\hat{c}_t$, then the production is determined by:

$$\hat{y}_t=\mathbb{E}_t\left\{\hat{y}_{t+1}\right\}-\frac{1}{\sigma}\left(\hat{r}_t-\mathbb{E}_t\left\{\widehat{\pi}_{t+1}\right\}\right)+\varepsilon_t^D \tag{33}$$

The labour supply equation in log-deviation is:

$$\hat{w}_t=(\sigma+\varphi)\,\hat{y}_t \tag{34}$$

where $\hat{w}_t$ denotes the variations of the real wage.

Regarding the new keynesian Phillips curve, we apply a first order approximation:

$$\mathbb{E}_t\sum_{\tau=0}^{+\infty}\frac{\lambda_{t+\tau}^c}{\lambda_t^c}\left(\beta\theta\right)^\tau\frac{P_{it}^*}{P_{t+\tau}}Y_{it+\tau}-\mathbb{E}_t\sum_{\tau=0}^{+\infty}\frac{\lambda_{t+\tau}^c}{\lambda_t^c}\left(\beta\theta\right)^\tau e^{\varkappa\varepsilon_t^S}\frac{\epsilon}{\epsilon-1}mc_{it+\tau}Y_{it+\tau}=0$$

$$\mathbb{E}_t\sum_{\tau=0}^{+\infty}(\beta\theta)^\tau\bar{Y}\left(\hat{y}_{it+\tau}+\Delta\hat{\lambda}_{t+\tau}^c+\hat{p}_{it}^*-\hat{p}_{t+\tau}\right)-\mathbb{E}_t\sum_{\tau=0}^{+\infty}(\beta\theta)^\tau\frac{\epsilon}{\epsilon-1}\overline{mc}\bar{Y}\left(\hat{y}_{it+\tau}+\Delta\hat{\lambda}_{t+\tau}^c+\widehat{mc}_{it+\tau}+\varkappa\varepsilon_{t+\tau}^S\right)=0$$

Since $\overline{mc}\epsilon/(\epsilon-1)=1$, we can gather back element into the same sum:

$$\mathbb{E}_t\sum_{\tau=0}^{+\infty}(\beta\theta)^\tau\left[\left(\hat{y}_{it+\tau}+\Delta\hat{\lambda}_{t+\tau}^c+\hat{p}_{it}^*-\hat{p}_{t+\tau}\right)-\left(\hat{y}_{it+\tau}+\Delta\hat{\lambda}_{t+\tau}^c+\widehat{mc}_{it+\tau}+\varkappa\varepsilon_{t+\tau}^S\right)\right]=0$$

$$\mathbb{E}_t\sum_{\tau=0}^{+\infty}(\beta\theta)^\tau\left[\left(\hat{p}_{it}^*-\hat{p}_{t+\tau}\right)-\left(\widehat{mc}_{it+\tau}+\varkappa\varepsilon_{t+\tau}^S\right)\right]=0$$

Recall that $\sum_s^{+\infty}a^s=1/(1-a^s)$, the expression for $\hat{p}_{it}^*$ (which is independent of the sum) can be rewritten as:

$$\frac{\hat{p}_{it}^*}{1-\beta\theta}=\mathbb{E}_t\sum_{\tau=0}^{+\infty}(\beta\theta)^\tau\left[\hat{p}_{t+\tau}+\widehat{mc}_{it+\tau}+\varkappa\varepsilon_{t+\tau}^S\right]$$

Since $x_t=\sum_s^{+\infty}a^sy_{t+s}=y_t+ax_{t+1}$,we can get rid of the sum:

$$\frac{\hat{p}_{it}^*}{1-\beta\theta}=\hat{p}_t+\widehat{mc}_{it}+\varkappa\varepsilon_t^S+\frac{\beta\theta}{1-\beta\theta}\hat{p}_{it+1}^*$$

$$\hat{p}_{it}^*=(1-\beta\theta)\left(\hat{p}_t+\widehat{mc}_{it}+\varkappa\varepsilon_t^S\right)+\beta\theta\hat{p}_{it+1}^*$$

Up to a first order approximation of the firm price optimization solution and the aggregate price equation:

$$\hat{p}_t=\theta\hat{p}_{t-1}+(1-\theta)\hat{p}_t^*$$

$$\hat{p}_t^*=\frac{\hat{p}_t-\theta\hat{p}_{t-1}}{(1-\theta)}$$

We can substitute the optimal price $\hat{p}_t^*$:

$$\frac{\hat{p}_t-\theta\hat{p}_{t-1}}{(1-\theta)}=(1-\beta\theta)\left(\hat{p}_t+\widehat{mc}_t+\varkappa\varepsilon_t^S\right)+\beta\theta\left[\frac{\hat{p}_{t+1}-\theta\hat{p}_t}{(1-\theta)}\right]$$

$$\hat{p}_t-\theta\hat{p}_{t-1}=(1-\theta)\left(1-\beta\theta\right)\left(\hat{p}_t+\widehat{mc}_t+\varkappa\varepsilon_t^S\right)+\beta\theta\left[\hat{p}_{t+1}-\theta\hat{p}_t\right]$$

To express in terms of inflation, we exploit, $\hat{p}_{t-1} = \hat{p}_t - \hat{\pi}_t$, and include $\hat{p}_t - \hat{p}_t$ next to $\hat{p}_{t+1}$ to obtain the forward inflation rate:

$$
\begin{aligned}
\hat{p}_t - \theta\left(\hat{p}_t - \hat{\pi}_t\right) &= (1-\theta)(1-\beta\theta)\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\theta\left[\hat{p}_{t+1} + \hat{p}_t - \hat{p}_t - \theta\hat{p}_t\right] \\
\hat{p}_t - \theta\left(\hat{p}_t - \hat{\pi}_t\right) &= (1-\theta)(1-\beta\theta)\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\theta\left[\hat{\pi}_{t+1} + (1-\theta)\hat{p}_t\right] \\
\frac{1}{\theta}\hat{p}_t - \hat{p}_t + \hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\left[\hat{\pi}_{t+1} + (1-\theta)\hat{p}_t\right] \\
\left(\frac{1-\theta}{\theta}\right)\hat{p}_t + \hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\hat{\pi}_{t+1} + \beta(1-\theta)\hat{p}_t \\
\hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\hat{\pi}_{t+1} + (1-\theta)\left[\beta - \frac{1}{\theta}\right]\hat{p}_t \\
\hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\hat{\pi}_{t+1} + (1-\theta)\left[\frac{\beta\theta - 1}{\theta}\right]\hat{p}_t \\
\hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\hat{p}_t + \widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\hat{\pi}_{t+1} - \frac{(1-\theta)(1-\beta\theta)}{\theta}\hat{p}_t \\
\hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\left(\widehat{mc}_t + \varkappa\varepsilon_t^S\right) + \beta\hat{\pi}_{t+1} \\
\hat{\pi}_t &= \frac{(1-\theta)(1-\beta\theta)}{\theta}\widehat{mc}_t + \beta\hat{\pi}_{t+1} + \varepsilon_t^S
\end{aligned}
$$

with $\varkappa$ equal to the inverse to the slope of the new keynesian phillips curve to normalize the shock to one.

With $\widehat{mc}_t = \widehat{w}_t$ and linearizing the production function, $y_t = h_t$, and the labour supply equation:

$$\hat{w}_t = (\sigma + \varphi)\, y_t,$$

then the log Phillips curve can (finally) be rewritten through its very well known expression:

$$\widehat{\pi}_t = \beta\mathbb{E}_t\widehat{\pi}_{t+1} + \frac{(1-\theta)(1-\theta\beta)}{\theta}(\sigma+\varphi)\hat{y}_t + \varepsilon_t^S. \tag{35}$$

where we can the auxialliary parameter $\kappa = (1-\theta)(1-\theta\beta)/\theta$. Finally, the monetary policy is determined by:

$$\hat{r}_t = \rho\hat{r}_{t-1} + (1-\rho)\left(\phi^\pi\widehat{\pi}_t + \phi^y\hat{y}_t\right) + \varepsilon_t^R. \tag{36}$$